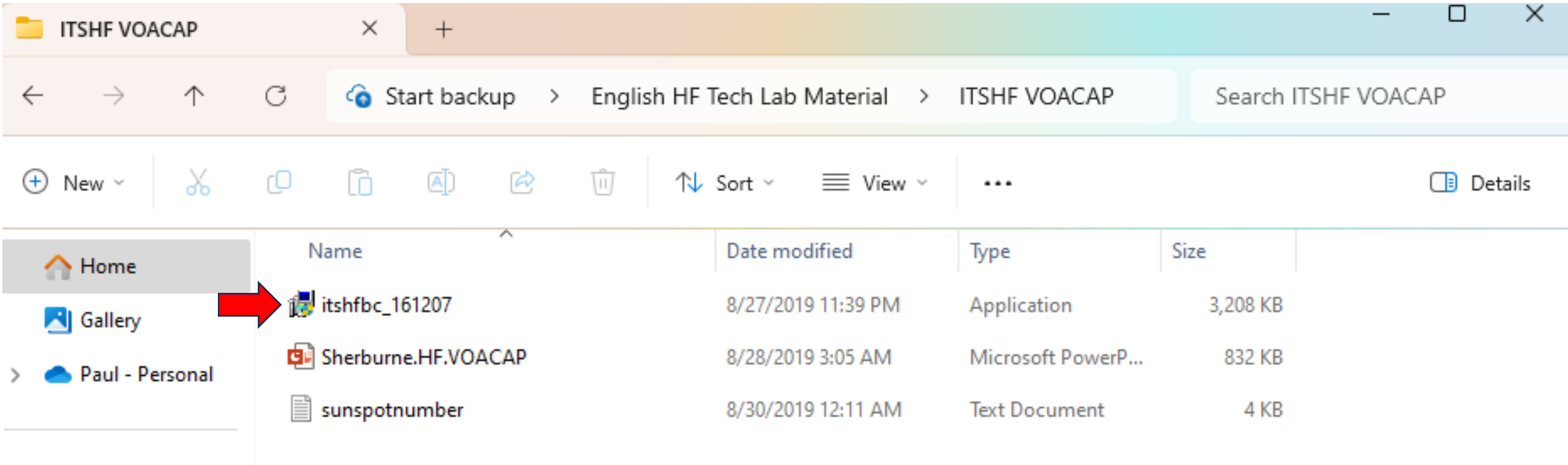


VOACAP HF Skills



Using the provided program file, install the program per normal software installation instructions

If you created a desktop shortcut, click the icon to open VOACAP



	HFant	3/2/2021 8:12 PM	Shortcut	2 KB
	ICEAREA INVERSE	3/2/2021 8:12 PM	Shortcut	2 KB
	ICEAREA	3/2/2021 8:12 PM	Shortcut	2 KB
	ICEPAC	3/2/2021 8:12 PM	Shortcut	2 KB
	News	3/2/2021 8:12 PM	Shortcut	2 KB
	readme.txt	3/2/2021 8:12 PM	Shortcut	1 KB
	REC533	3/2/2021 8:12 PM	Shortcut	2 KB
	RECAREA	3/2/2021 8:12 PM	Shortcut	2 KB
	S_I ICEPAC	3/2/2021 8:12 PM	Shortcut	2 KB
	S_I VOACAP	3/2/2021 8:12 PM	Shortcut	2 KB
	Uninstall ITS HF Propagation	3/2/2021 8:12 PM	Shortcut	2 KB
	VOAAREA INVERSE	3/2/2021 8:12 PM	Shortcut	2 KB
	VOAAREA	3/2/2021 8:12 PM	Shortcut	2 KB
	VOACAP	3/2/2021 8:12 PM	Shortcut	2 KB



There are several options, please select and click “VOACAP”

Click "Method"



VOACAP Point-to-Point data input - Version 16.1207W

File Run View Save to: Help

Method 2 = Ionograms

Year 2024 **Coefficients** CCIR (Oslo)

Time 01 to 24 by 1 hours UT

Groups Month.Day= 2.01
SSN = 113

Transmitter 40.58N 105.08W FORT COLLINS **Swap Tx-Rx**

Receiver 30.95N 97.53W SALADO

Path Short Distances: 1268km 685nmi 788mi Azimuth: 145.3deg

Freq(MHz) 6.075 7.200 9.700 11.850 13.700 15.350 17.725 21.650 25.885

System

Noise	Min Angle	Req.Rel.	Req SNR	Multi Tol	Multi Del	Absorp
145 (-dBw)	3.00deg	90%	73dB	3.00dB	0.10msec	Normal

Eprob 1.00*foE 1.00*foF1 1.00*foF2 0.00*foEs

Tx Antenna

#	Min	Max	Design	Directory\Filename.sfx	Model	MainBeam	Power kW
1	2	30	0.000	default \Isotrope	+ 0.0 dBi	132.0	0.0200

Rx Antenna default \Isotrope 0.0deg 0.00dB

Input Help:

Change propagation METHOD



Select the Propagation METHOD to use:

- 1 = Ionospheric parameters
- 2 = Ionograms
- 3 = MUF-FOT lines (nomogram)
- 4 = MUF-FOT graph (use 11 or 28)
- 5 = HPF-MUF-FOT graph
- 6 = MUF-FOT-Es graph (use 11)
- 7 = FOT-MUF table (full ionosphere)
- 8 = MUF-FOT graph (use 11 or 28)
- 9 = HPF-MUF-FOT graph
- 10 = MUF-FOT-ANG graph
- 11 = MUF-FOT-Es graph - real graph, not line printer
- 12 = MUF by magnetic indices, K(not implemented)
- 13 = Transmitter antenna pattern
- 14 = Receiver antenna pattern
- 15 = Both transmitter & receiver antenna patterns
- 16 = System performance (S.P.)
- 17 = Condensed system performance, reliability
- 18 = Condensed system performance, service probability
- 19 = Propagation path geometry
- 20 = Complete system performance (C.S.P.)
- 21 = Forced long path model (C.S.P.)
- 22 = Forced short path model (C.S.P.)
- 23 = User selected output (set by TOPLINES & BOTLINES)
- 24 = MUF-REL table
- 25 = All modes table
- 26 = MUF-LUF-FOT table (nomogram)
- 27 = FOT-LUF graph (use 28)
- 28 = MUF-FOT-LUF graph - real graph, not line printer
- 29 = MUF-LUF graph (use 28)
- 30 = Short/Long smoothing (7-10000 km) - Recommended



1. Select "20 complete system performance"

2. Then "Accept"

1. Set the year
2. Set the time for 24 hours
3. Set Coefficients to "CCIR (Oslo)"



VOACAP Point-to-Point data input - Version 16.1207W

File Run View Save to: Help

Method	20 = Complete system performance (C.S.P.)	
Year	2024	Coefficients CCIR (Oslo)
Time	01 to 24 by 1 hours	UT
Groups	Month.Day= 2.01 SSN = 113	
Transmitter	40.58N 105.08W FORT COLLINS	
Receiver	30.95N 97.53W SALADO	
Path	Short Distances: 1268km 685nmi 788mi Azimuth: 145.3deg	
Freq(MHz)	6.075 7.200 9.700 11.850 13.700 15.350 17.725 21.650 25.885	
System	Noise Min Angle Req.Rel. Req SNR Multi Tol Multi Del Absorp 145 (-dBw) 3.00deg 90% 73dB 3.00dB 0.10msec Normal	
Eprob	1.00*foE 1.00*foF1 1.00*foF2 0.00*foEs	
Tx Antenna	# Min Max Design Directory\Filename.sfx Model MainBeam Power kW 1 2 30 0.000 default \Isotrope + 0.0 dBi 132.0 0.0200	
Rx Antenna	default \Isotrope 0.0deg 0.00dB	

Input Help:

4. Click "Groups" and go to the Next screen

On this screen, hover your mouse over each button and field to see a help screen descriptor.

In the first fillable column, 2.01 represents Month.day or February 1st.

3.31 would represent March 31st, etc.

The second blank field is for the predicted Smoothed Sunspot Number.

If you have access to the Internet, you can google SSN.

If you do not have Internet access, listen to the Space weather report on WWV at 18 past the hour or WWVH at 45 past the hour and record the Solar Flux Index (SFI)

Go to next screen

Change MONTH/SSN parameters

AcceptCancelJanuarySeasonsAll MONTHs

MonthsSSNs

1	2.01	113
2	0.00	0
3	0.00	0
4	0.00	0
5	0.00	0
6	0.00	0
7	0.00	0
8	0.00	0
9	0.00	0
10	0.00	0

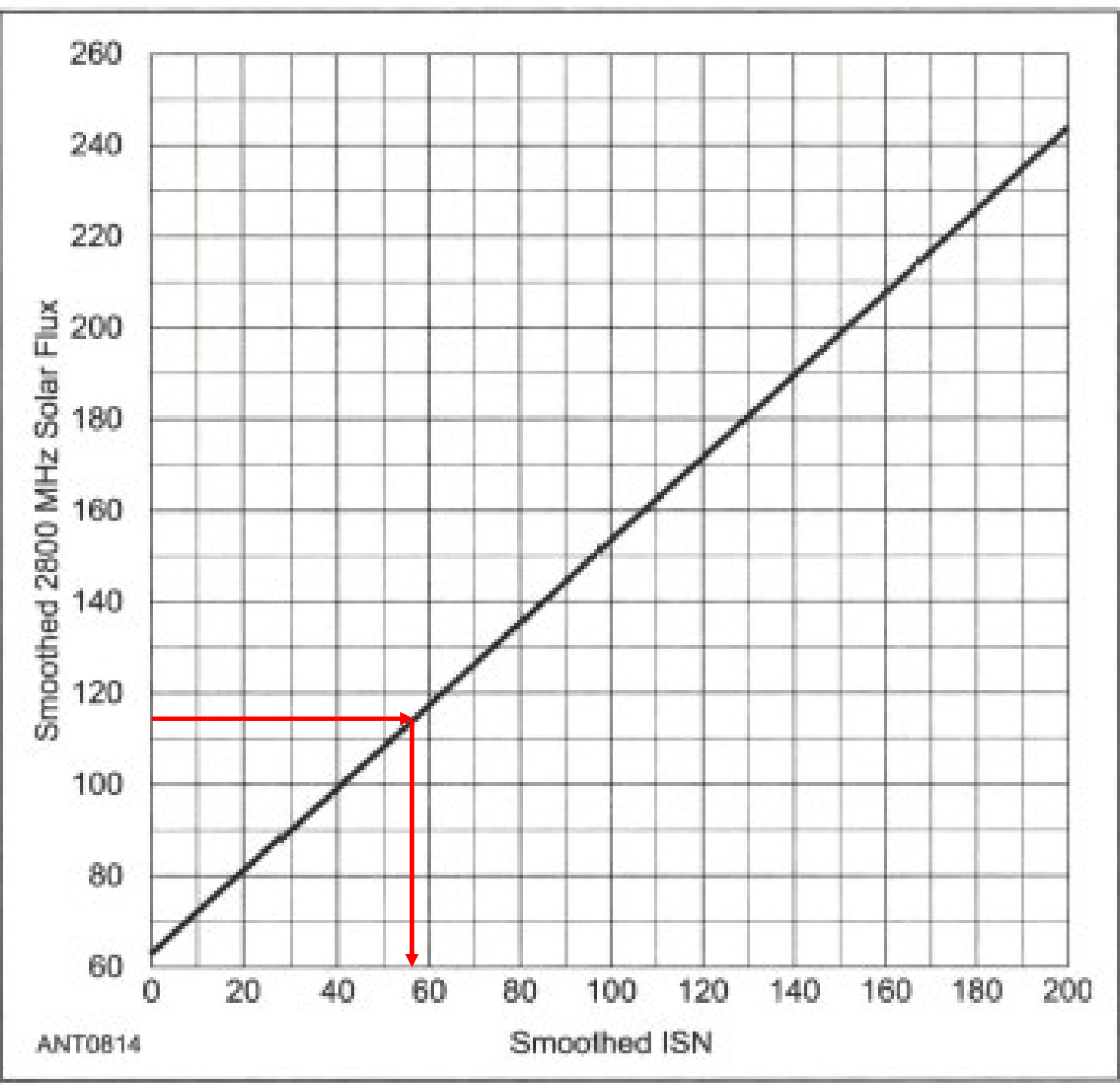
Input Help:

How to approximate the Smoothed Sunspot Number when you know the Solar Flux Index (SFI)

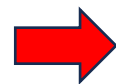
Monitor WWV at 18 mins past the hour or WWVH at 45 mins past the hour, record the SFI

Find the SFI on the vertical axis, move straight across to intersect the sloping black line then drop straight down to read the approximate Smoothed Sunspot Number (SSN).

Enter this SSN in the VOACAP groups field



Click "Transmitter" to set the transmitter parameters



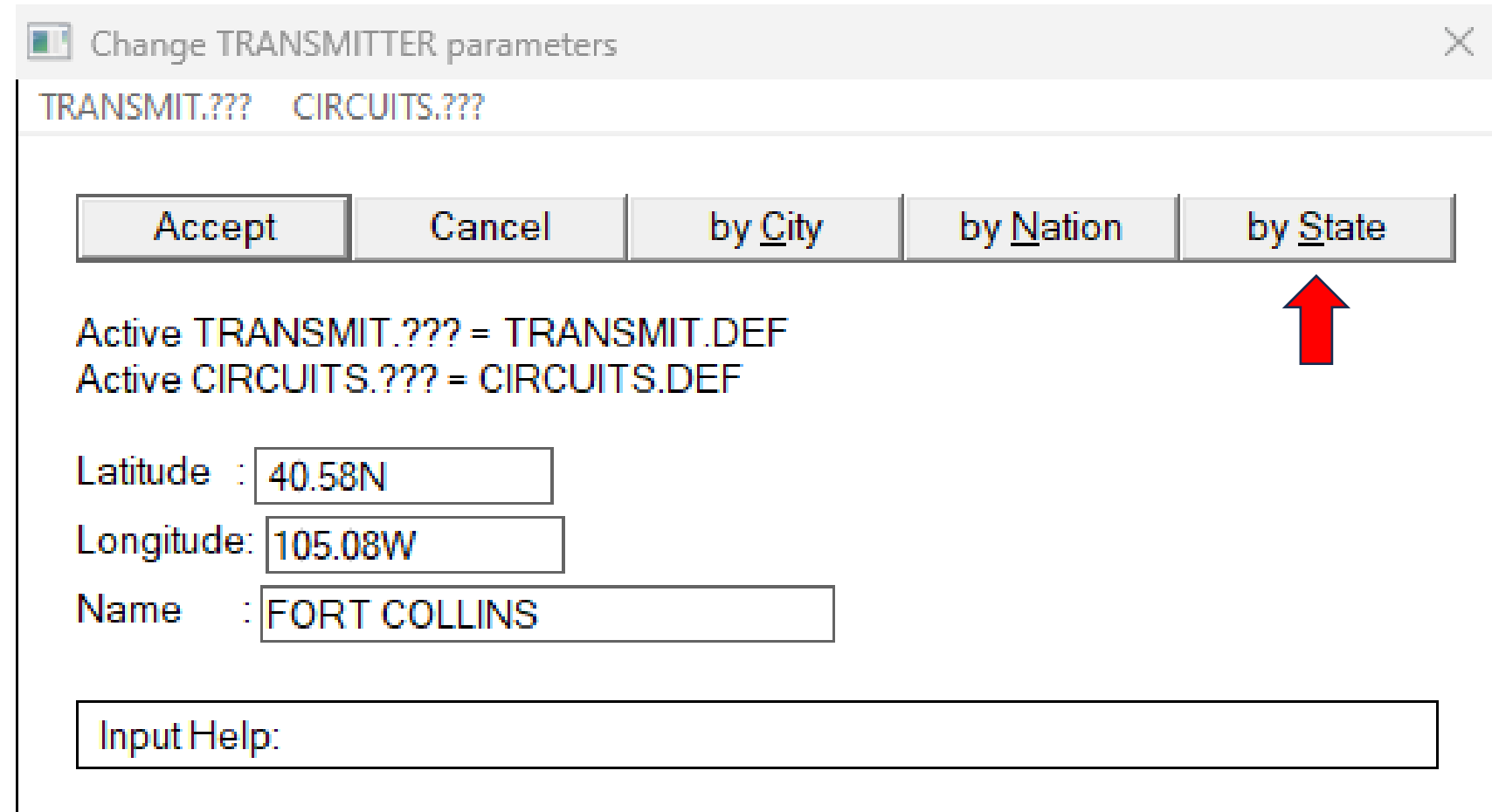
VOACAP Point-to-Point data input - Version 16.1207W

File Run View Save to: Help

Method	20 = Complete system performance (C.S.P.)									
Year	2024	Coefficients		CCIR (Oslo)						
Time	01 to 24 by 1 hours UT									
Groups	Month.Day= 2.01 SSN = 113									
Transmitter	40.58N 105.08W FORT COLLINS Swap Tx-Rx									
Receiver	30.95N 97.53W SALADO									
Path	Short Distances: 1268km 685nmi 788mi Azimuth: 145.3deg									
Freq(MHz)	6.075 7.200 9.700 11.850 13.700 15.350 17.725 21.650 25.885									
System	Noise	Min Angle	Req.Rel.	Req SNR	Multi Tol	Multi Del	Absorp			
	145 (-dBw)	3.00deg	90%	73dB	3.00dB	0.10msec	Normal			
Eprob	1.00*foE	1.00*foF1	1.00*foF2	0.00*foEs						
Tx Antenna	# Min Max Design Directory\Filename.sfx Model MainBeam Power kW									
	1	2	30	0.000	default	\Isotrope	+	0.0 dBi	132.0 0.0200	
Rx Antenna	default \Isotrope 0.0deg 0.00dB									

Input Help:

Click “by State”

A screenshot of a software window titled "Change TRANSMITTER parameters". The window has a standard Windows-style title bar with a close button (X) in the top right corner. Below the title bar, there is a header area with the text "TRANSMIT.???" and "CIRCUITS.???" in a monospaced font. Below this header is a row of five buttons: "Accept", "Cancel", "by City", "by Nation", and "by State". A red arrow points upwards to the "by State" button. Below the buttons, there are two lines of text: "Active TRANSMIT.???" followed by "= TRANSMIT.DEF" and "Active CIRCUITS.???" followed by "= CIRCUITS.DEF". Below these lines are three input fields: "Latitude : 40.58N", "Longitude: 105.08W", and "Name : FORT COLLINS". At the bottom of the window is a large text area labeled "Input Help:".

TRANSMIT.???

CIRCUITS.???

AcceptCancelby Cityby Nationby State

Active TRANSMIT.???

= TRANSMIT.DEF

Active CIRCUITS.???

= CIRCUITS.DEF

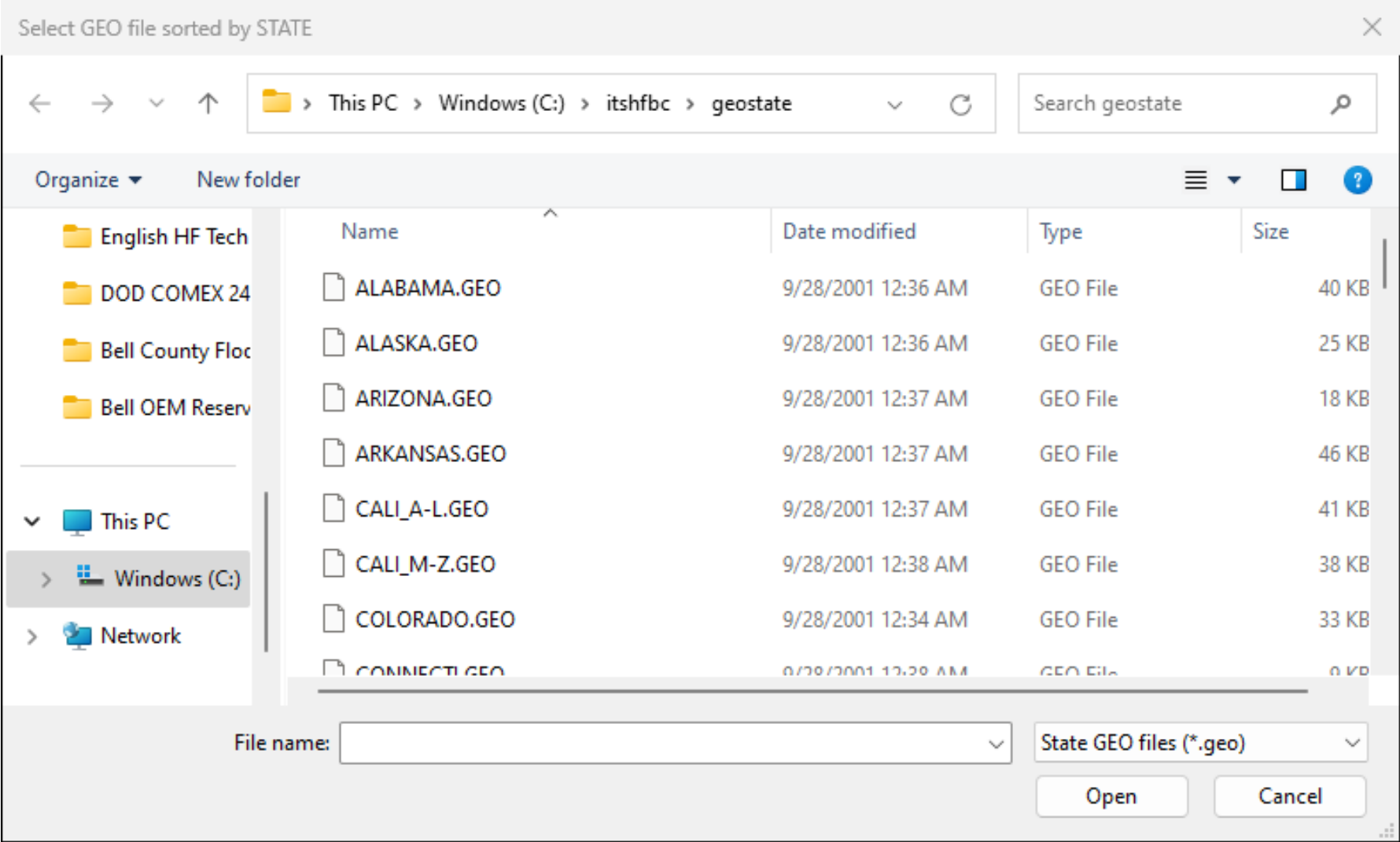
Latitude : 40.58N

Longitude: 105.08W

Name : FORT COLLINS

Input Help:

Find the correct state then
double click the state to select
the nearest city



Scroll through the selection of cities,
highlight and accept the correct city

Select coordinates from:C:\itshfbc\geostate\COLORADO.GEO					Exit Accept	
=====CITY=====		=====State		Latitude	Longitude	
ADAMS CITY	CO	39	50	N	104	56 W
AGATE	CO	39	28	N	103	57 W
AGUILAR	CO	37	42	N	104	39 W
AIR FORCE ACADEMY	CO	39	0	N	104	52 W
AKRON	CO	40	10	N	103	13 W
ALAMOSA	CO	37	28	N	105	52 W
ALAMOSA EAST	CO	37	29	N	105	51 W
ALLENSPARK	CO	40	12	N	105	32 W
ALLISON	CO	37	2	N	107	29 W
ALMA	CO	39	17	N	106	4 W
AMHERST	CO	40	41	N	102	10 W
ANDRIX	CO	37	17	N	103	12 W
ANTON	CO	39	45	N	103	13 W
ANTONITO	CO	37	5	N	106	1 W
APPLEWOOD	CO	39	44	N	105	11 W
ARAPAHOE	CO	38	51	N	102	11 W
ARLINGTON	CO	38	20	N	103	21 W
AROYA	CO	38	52	N	103	8 W
ARRIBA	CO	39	17	N	103	17 W
ARTMUS	CO	39	44	N	104	59 W
ARVADA	CO	39	48	N	105	5 W
ASPEN	CO	39	11	N	106	49 W
ATWOOD	CO	40	33	N	103	16 W
AULT	CO	39	44	N	104	52 W
AURORA	CO	39	44	N	104	52 W
AVON	CO	39	38	N	106	31 W
AVONDALE	CO	38	14	N	104	21 W
AXIAL	CO	40	17	N	107	47 W
BAILEY	CO	39	25	N	105	29 W
BASALT	CO	39	22	N	107	2 W

1. Do the same process to set the receiver location as you did to select the Transmitter location

Note: Once you've selected both transmitter and receiver locations, VOACAP now displays the azimuth and distance between the selected locations

2. Click "Swap Tx-Rx" this gives you the back azimuth between the two locations

VOACAP Point-to-Point data input - Version 16.1207W

File Run View Save to: Help

Method	20 = Complete system performance (C.S.P.)									
Year	2024	Coefficients		CCIR (Oslo)						
Time	01 to 24 by 1 hours UT									
Groups	Month.Day= 2.01 SSN = 113									
Transmitter	40.58N 105.08W FORT COLLINS									
Receiver	30.95N 97.53W SALADO									
Path	Short	Distances: 1268km 685nmi 788mi Azimuth: 145.3deg								
Freq(MHz)	6.075 7.200 9.700 11.850 13.700 15.350 17.725 21.650 25.885									
System	Noise	Min Angle	Req.Rel.	Req SNR	Multi Tol	Multi Del	Absorp			
	145 (-dBw)	3.00deg	90%	73dB	3.00dB	0.10msec	Normal			
Eprob	1.00*foE	1.00*foF1	1.00*foF2	0.00*foEs						
Tx Antenna	# Min Max Design Directory\Filename.sfx Model MainBeam Power kW									
	1	2	30	0.000	default	\Isotrope	+	0.0 dBi	132.0	0.0200
Rx Antenna	default \Isotrope 0.0deg 0.00dB									
Input Help:										

Note: The azimuths given in VOACAP are true north. To get a magnetic azimuth, you will need to adjust for the declination angle

1. Click “Tx Antenna” and go to the next screen to set TX antenna and power output parameters



2. Select “Rx Antenna” and set to “default/isotrope”



VOACAP Point-to-Point data input - Version 16.1207W

File Run View Save to: Help

Method	20 = Complete system performance (C.S.P.)									
Year	2024	Coefficients		CCIR (Oslo)						
Time	01 to 24 by 1 hours UT									
Groups	Month.Day= 2.01 SSN = 113									
Transmitter	40.58N 105.08W FORT COLLINS									
Receiver	30.95N 97.53W SALADO									
Path	Short Distances: 1268km 685nmi 788mi Azimuth: 145.3deg									
Freq(MHz)	6.075 7.200 9.700 11.850 13.700 15.350 17.725 21.650 25.885									
System	Noise	Min Angle	Req.Rel.	Req SNR	Multi Tol	Multi Del	Absorp			
	145 (-dBw)	3.00deg	90%	73dB	3.00dB	0.10msec	Normal			
Eprob	1.00*foE	1.00*foF1	1.00*foF2	0.00*foEs						
Tx Antenna	# Min Max Design Directory\Filename.sfx Model MainBeam Power kW									
	1	2	30	0.000	default \Isotrope	+	0.0 dBi	132.0	0.0200	
Rx Antenna	default \Isotrope 0.0deg 0.00dB									
Input Help:										

Note the transmitter power is in KW;

WWV power is 10 Kw

A 100 W station would be 0.1 Kw

A manpack PRC-160 would be .02 Kw

Select “default/isotrop” for both xmt and rcv ants

Change TRANSMIT antenna parameters for VOACAP

AcceptCancel

Min	Max	Design	MainBeam		TxPower
02	30	0	131.9882 deg	at Rx	0.02 kW
		TxAnt=	default\Isotrope (+ 0.0 dBi)		
Unused	N/A	0	0 deg	at Rx	0 kW
		TxAnt=			
Unused	N/A	0	0 deg	at Rx	0 kW
		TxAnt=			
Unused	N/A	0	0 deg	at Rx	0 kW
		TxAnt=			

Input Help:

There is no need to change parameter settings for Freq, system or Fprob

VOACAP Point-to-Point data input - Version 16.1207W

File Run View Save to: Help

Method

20 = Complete system performance (C.S.P.)

Year

2024

Coefficients

CCIR (Oslo)

Time

01 to 24 by 1 hours UT

Groups

Month.Day= 2.01

SSN = 113

Transmitter

40.58N 105.08W FORT COLLINS

Swap Tx-Rx

Receiver

30.95N 97.53W SALADO

Path

Short Distances: 1268km 685nmi 788mi Azimuth: 145.3deg

Freq(MHz)

6.075 7.200 9.700 11.850 13.700 15.350 17.725 21.650 25.885

System

Noise Min Angle Req.Rel. Req SNR Multi Tol Multi Del Absorp

145 (-dBw) 3.00deg 90% 73dB 3.00dB 0.10msec Normal

Fprob

1.00*foE 1.00*foF1 1.00*foF2 0.00*foEs

Tx Antenna

Min Max Design Directory\Filename.sfx Model MainBeam Power kW

1 2 30 0.000 default \Isotrope + 0.0 dBi 132.0 0.0200

Rx Antenna

default \Isotrope 0.0deg 0.00dB

Input Help:

Once you've correctly set all the parameters, click "Run" then "Graph"

You'll see an interim screen pop up then disappear quickly

VOACAP Point-to-Point data input - Version 16.1207W

Run View Save to: Help

Method 20 = Complete system performance (C.S.P.)

Year 2024 **Coefficients** CCIR (Oslo)

Time 01 to 24 by 1 hours UT

Groups Month.Day= 2.01
SSN = 113

Transmitter 40.58N 105.08W FORT COLLINS **Swap Tx-Rx**

Receiver 30.95N 97.53W SALADO

Path Short Distances: 1268km 685nmi 788mi Azimuth: 145.3deg

Freq(MHz) 6.075 7.200 9.700 11.850 13.700 15.350 17.725 21.650 25.885

System

Noise	Min Angle	Req.Rel.	Req SNR	Multi Tol	Multi Del	Absorp
145 (-dBw)	3.00deg	90%	73dB	3.00dB	0.10msec	Normal

Eprob 1.00*foE 1.00*foF1 1.00*foF2 0.00*foEs

Tx Antenna

#	Min	Max	Design	Directory\Filename.sfx	Model	MainBeam	Power kW
1	2	30	0.000	default \Isotrope	+ 0.0 dBi	132.0	0.0200

Rx Antenna default \Isotrope 0.0deg 0.00dB

Input Help:

This is the next screen you'll see. From here, you can select any of these options.

Please click "TANGLE" short "For take off angle"

Select PARAMETER to plot from: Group # 1 2.01 113ssn

Exit

Group # 1 2.01 113ssn

Parameter	[Minimum/Maximum]
TANGLE= Radiation angle (degrees)	[4.20/ 30.20]
DELAY = Time delay (milliseconds)	[4.30/ 5.20]
VHTE = Virtual height (km)	[79.00/ 426.00]
MUFday= % of days/month sky-wave propagation expected at MUF	[0.00/ 100.00]
LOSS = Median system loss (dB)	[112.00/ 261.00]
DBU = Median field strength at receive location (dBu)	[-135.00/ 19.00]
SDBW = Median signal power at receiver (dBW)	[-248.00/ -99.00]
NDBW = Median noise power at receiver (dBW)	[-173.00/ -136.00]
SNR = Median signal-to-noise ratio (dB)	[-108.00/ 48.00]
RPWRG = Required power & antenna gain to achieve reliab (dB)	[38.00/ 194.00]
REL = Time availability, % time SNR exceeds required SNR	[0.00/ 0.00]
MPROB = Probability additional mode in multipath tolerances	[0.00/ 0.00]
SPRB = Service probability, required reliability will be met	[0.00/ 4.00]
SIGLW = Lower decile signal pwr (field strength & loss) (dB)	[3.70/ 25.00]
SIGUP = Upper decile signal pwr (field strength & loss) (dB)	[1.10/ 25.00]
SNRLW = Lower decile SNR increment (dB)	[9.20/ 26.80]
SNRUP = Upper decile SNR increment (dB)	[5.30/ 25.70]
TGAIN = Transmitter Antenna Gain (dB)	[0.00/ 0.00]
RGAIN = Receiver Antenna Gain (dB)	[0.00/ 0.00]
SNRxx = Signal-to-Noise ratio (dB) at Req. Rel.	[-121.00/ 35.00]



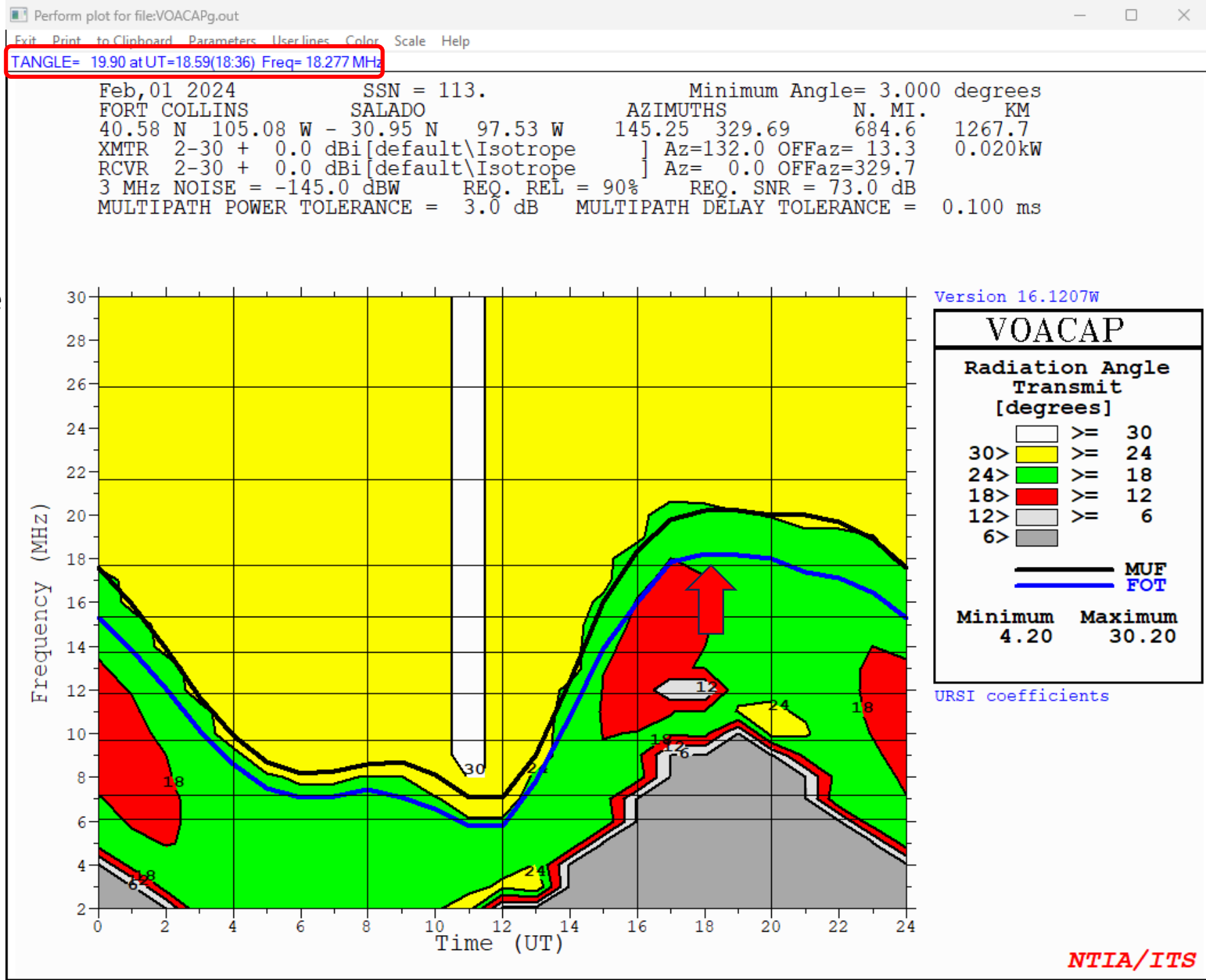
This is the TANGLE screen.

Note: the black line represents the predicted MUF freqs throughout the day.

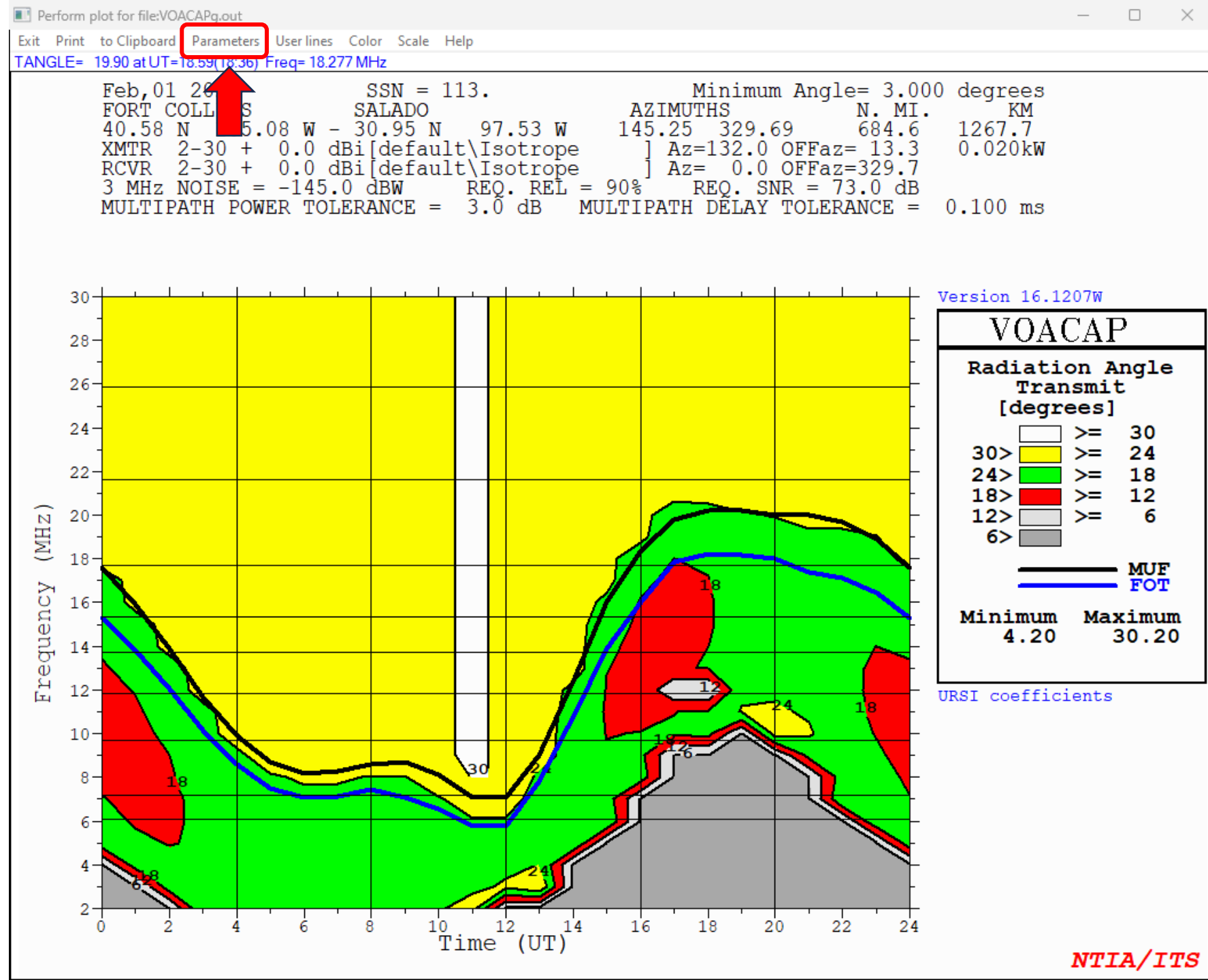
Note: the Blue line indicates the predicted frequency of optimum transmission throughout the day

Place the cursor anywhere on the blue line and click.

In the top, the screen will display the predicted TANGLE, time, and freq for where you clicked the cursor.



From this screen, you can simply
Click “parameters” to take you
back to the other parameters that
are available



Select PARAMETER to plot from: Group # 1 2.01 113ssn



Exit

Group # 1 2.01 113ssn

Parameter

[Minimum/Maximum]

TANGLE= Radiation angle (degrees)	[4.20/ 30.20]
DELAY = Time delay (milliseconds)	[4.30/ 5.20]
VHTE = Virtual height (km)	[79.00/ 426.00]
MUFday= % of days/month sky-wave propagation expected at MUF	[0.00/ 100.00]
LOSS = Median system loss (dB)	[112.00/ 261.00]
DBU = Median field strength at receive location (dBu)	[-135.00/ 19.00]
SDBW = Median signal power at receiver (dBW)	[-248.00/ -99.00]
NDBW = Median noise power at receiver (dBW)	[-173.00/ -136.00]
SNR = Median signal-to-noise ratio (dB)	[-108.00/ 48.00]
RPWRG = Required power & antenna gain to achieve reliab (dB)	[38.00/ 194.00]
REL = Time availability, % time SNR exceeds required SNR	[0.00/ 0.00]
MPROB = Probability additional mode in multipath tolerances	[0.00/ 0.00]
SPRB = Service probability, required reliability will be met	[0.00/ 4.00]
SIGLW = Lower decile signal pwr (field strength & loss) (dB)	[3.70/ 25.00]
SIGUP = Upper decile signal pwr (field strength & loss) (dB)	[1.10/ 25.00]
SNRLW = Lower decile SNR increment (dB)	[9.20/ 26.80]
SNRUP = Upper decile SNR increment (dB)	[5.30/ 25.70]
TGAIN = Transmitter Antenna Gain (dB)	[0.00/ 0.00]
RGAIN = Receiver Antenna Gain (dB)	[0.00/ 0.00]
SNRxx = Signal-to-Noise ratio (dB) at Req. Rel.	[-121.00/ 35.00]